BUILD-YOUR-OWN-UNO
Assembly & User Documentation

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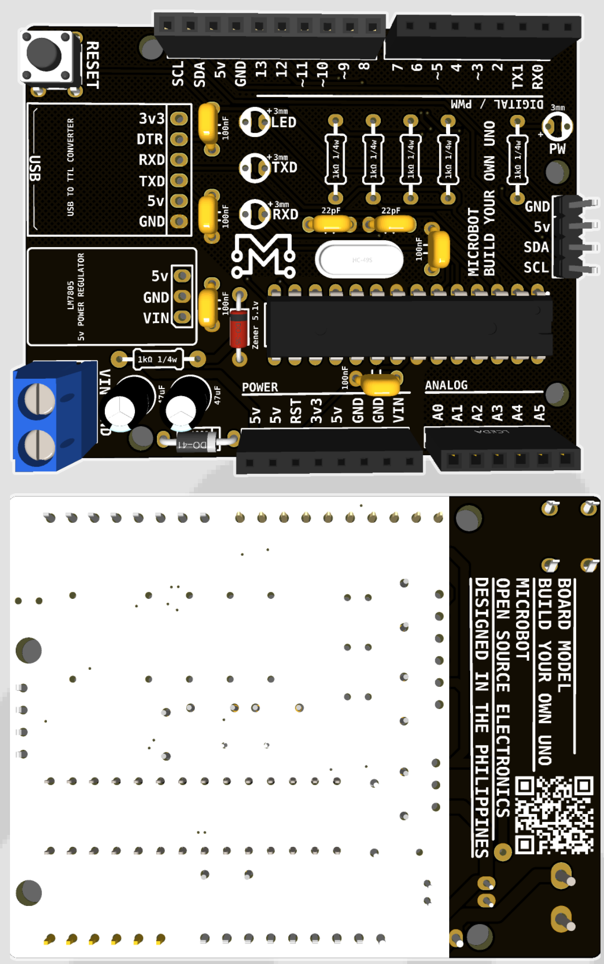
OVERVIEW:

The Build-Your-Own Arduino Kit is a DIY microcontroller board designed for students, hobbyists, and electronics enthusiasts who want to learn both electronics assembly and microcontroller programming.

Unlike pre-assembled boards from Arduino, this kit ships unassembled so users can solder and construct the board themselves, gaining hands-on experience with embedded systems hardware. Once assembled, the board functions similarly to an Arduino-Uno-class microcontroller and can be programmed using the Arduino IDE.

Since this microcontroller meets all the criteria of compatibility with the Arduino IDE and Arduino Web Editor, programming the microcontroller follows the same procedure as any other Arduino UNO board.

BOARD LAYOUT:



(Board schematic available on page 3)

TECHNICAL SPECIFICATIONS:

Microcontroller	ATMEGA328P (DIP package)
Operating Voltage	5v
Input Voltage	USB or external 7V~15V DC input
Output voltage	5V 1A DC output 3.3V 6mA DC output
Clock speed	16 Mhz
Bootloader	Arduino Uno
Interface	USB-C
Digital I/O pins	14 (6 are PWM outputs)
Analog input pin	6
DC Current per I/O	40mA
Flash memory	32kb
SRAM	2kb
EEPROM	1kb
Dimensions	68.8mm x 53.4mm x 15mm
LED Pin	D13

PARTS LIST:

Part Name	Qty
Terminal Block	1
USB-TTL Converter	1
LM7805 Regulator	1
47uF Capacitor	2
1N4001 Diode	1
Tactile Switch	1
100nF Capacitor	1
1Kohm Resistor	6
LED	6
1x4 Male Header	1
1x6 Female Header	1
1x8 Female Header	1
1x10 Female Header	1
22nF Capacitor	2
16Mhz Oscillator	1
Zener Diode	1
ATMEGA328 IC	1
PCB	1

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Required Tools for Assembly:

- Soldering iron (25–40W recommended)
- Solder wire (lead-free or 60/40)
- Wire cutter
- Tweezers (optional)
- Multimeter (recommended for testing)

Assembly Guidelines:

1. Start with the smallest components
   > Resistors
   > Ceramic capacitors
   > Diodes
2. Install IC socket next
   > Ensure correct orientation (notch aligned)
3. Solder the crystal and supporting capacitors
4. Install the voltage regulator
5. Match flat side to PCB marking
6. Solder headers and connectors
7. Insert the ATMEGA328P-PU last
   > Align notch with socket marking

NOTE: *Do NOT power the board until all solder joints are inspected.*

First Power-Up Checklist:

- Before connecting USB or external power:
  > Inspect for solder bridges
  > Confirm regulator orientation
  > Verify capacitor polarity
  > Ensure IC is inserted correctly
- Connect USB cable
  > Check Power LED for light
  > Check Internal LED for Blink
  > Verify computer detects COM port

Programming the Board:

- Install Arduino IDE
- Install CH9340G drivers if needed
- Select board type:
  > Arduino Uno (ATmega328P)
- Select correct COM port
- Upload a simple Blink sketch
- If upload fails:
  > Check bootloader
  > Ensure correct COM port

Power Options:

Source	Supported
USB Power	YES
KF301 Terminal Block	YES
External 7–12V Input	RECOMMENDED

- Avoid supplying more than 12V to prevent over-heating.

Safety Notes:

- Do not power board while soldering
- Avoid reverse polarity connections
- Do not short 5V and 3V3 to GND
- Use proper ESD handling when touching ICs

Troubleshooting Quick Guide:

PROBLEM	POSSIBLE	FIX
No power LED	Regulator issue	Check soldering & polarity
Not detected by PC	Driver missing	Install CH9340G driver (link)
Upload fails	Bootloader missing	Burn bootloader via ISP (link)
Random resets	Power unstable	Check regulator heat & capacitors

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